Direct Multi-Horizon

GATv2 for Train Delay Prediction

Thinh Duy Do

Halicioglu Data Science Institute, UC San Diego

San Diego Undergraduate Tech Conference 2026

Why Train Delays Matter

Transit systems carry billions of trips each year

A primary delay spreads into knock-on delays

Propagation happens through shared track, dwell times, and headways

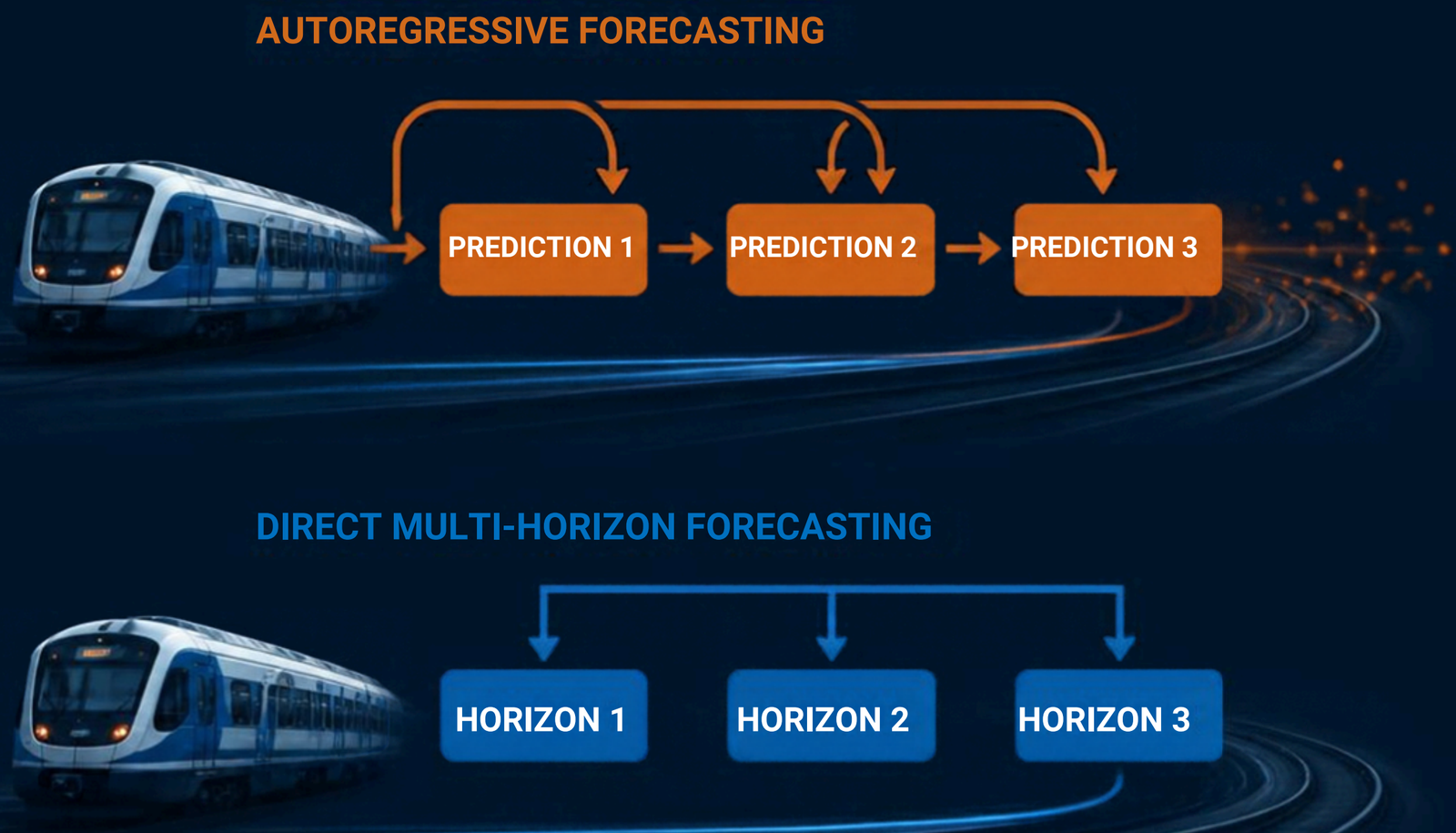
One delay can impact many

The Problem

Autoregressive forecasting reuses its own predictions, so errors compound.

Direct Multi-Horizon predicts all future events at once.

Is long-horizon error caused by the **graph model** or by the **decoding strategy**?



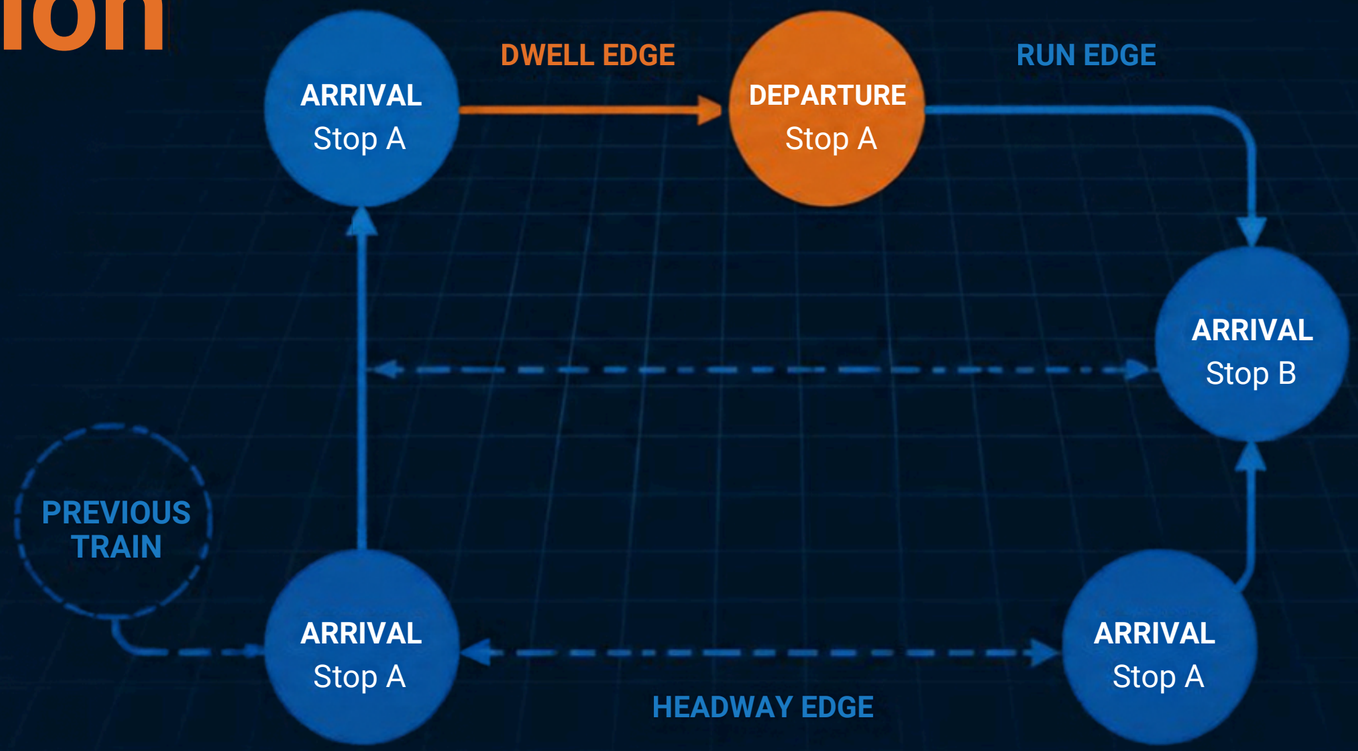
Data and Graph Construction

NYC Subway events are modeled as a graph

NYC Subway (subwaydata.nyc) matched with the MTA static GTFS schedule

Edges capture dwell, running, and headway relationships

Features include dwell and running times, trip progress, and time of day



DWELL

arrival → departure at the same stop

RUNNING

departure → next arrival of the same train

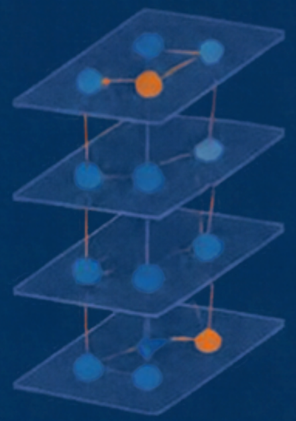
HEADWAY

consecutive trains on the same route, direction, and stop

Model Architecture

GRAPH INPUT

3-LAYER GATV2 ENCODER



TWO-STAGE HURDLE MODEL

CLASSIFY DELAY

Predict whether a delay occurs

REGRESS DELAY SIZE

Estimate the magnitude of the delay

MULTI-HORIZON HEADS

HORIZON t+1

HORIZON t+2

HORIZON t+3

OVERLAP COMBINATION

Inverse validation MAE weights

Overall Results

DMH Gatv2 achieves the best delay detection performance.

Graph models add value in detecting delays for operators.

DMH Gatv2

94.6%	93.8%	94.4%
Accuracy	Recall	F1

DMH Gatv2 lowers MAE vs Autoregressive Gatv2: **2.56** minutes vs **3.12** minutes

Persistence has the lowest MAE: **0.73** minutes

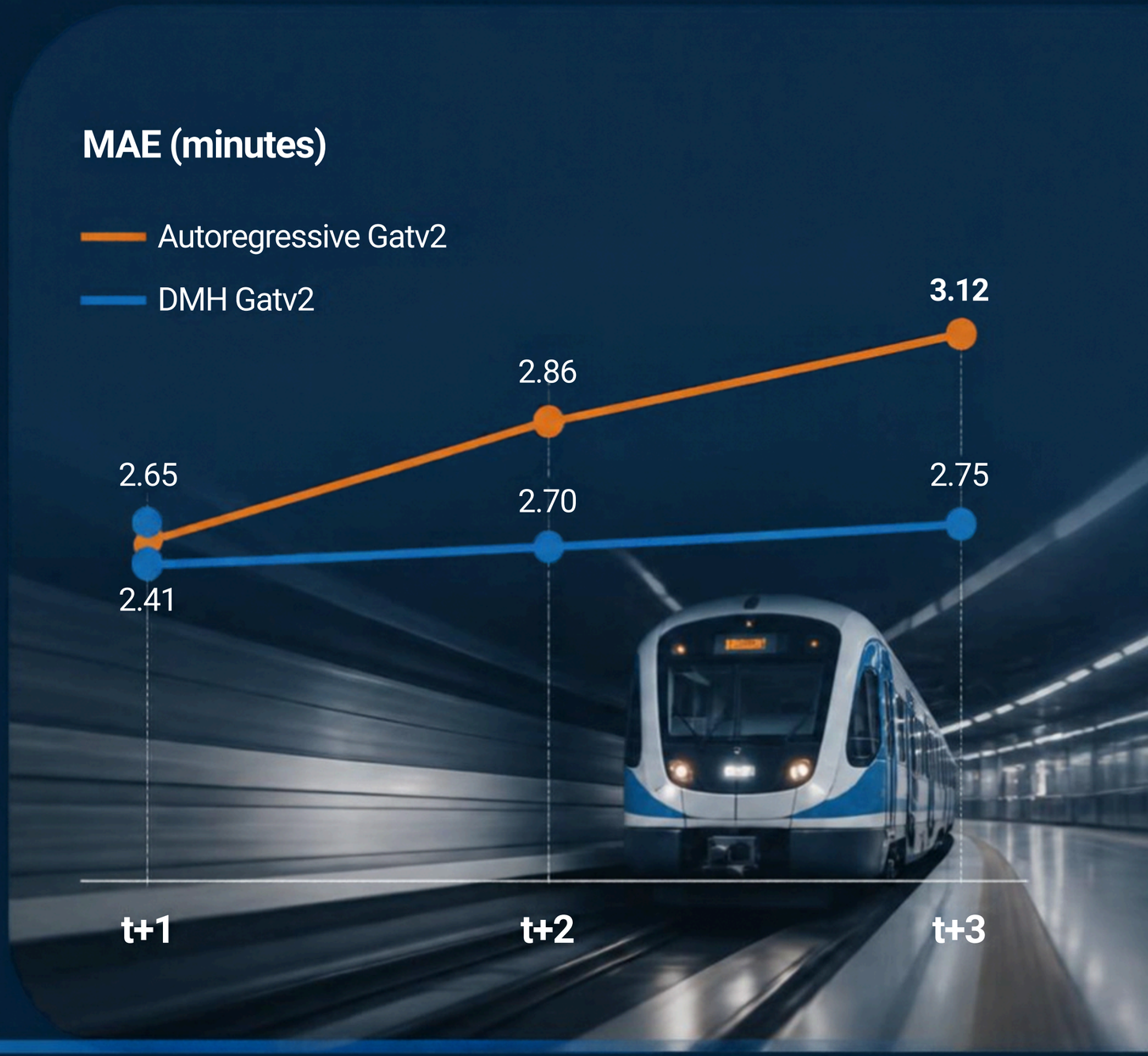
Key Finding
Stable Across Horizons

DMH Gatv2 keeps error nearly flat as we look further ahead

Autoregressive error compounds with each additional step

DMH Gatv2 delivers lower error at t+2 and t+3

Removing recursive decoding keeps performance stable



Limitations, Future Work and Takeaway

Current evaluation uses a single fixed horizon

Next, we wil explore additional horizons and more disruptive networks

Direct decoding cuts error accumulation while leveraging topology-aware modeling

Thank you!
Questions?